

pySFE: Stacking Fault Energy (SFE) Predictor

pySFE window

Stacking Fault Energy Predictor - Professional Edition

Alloy Composition (wt%)

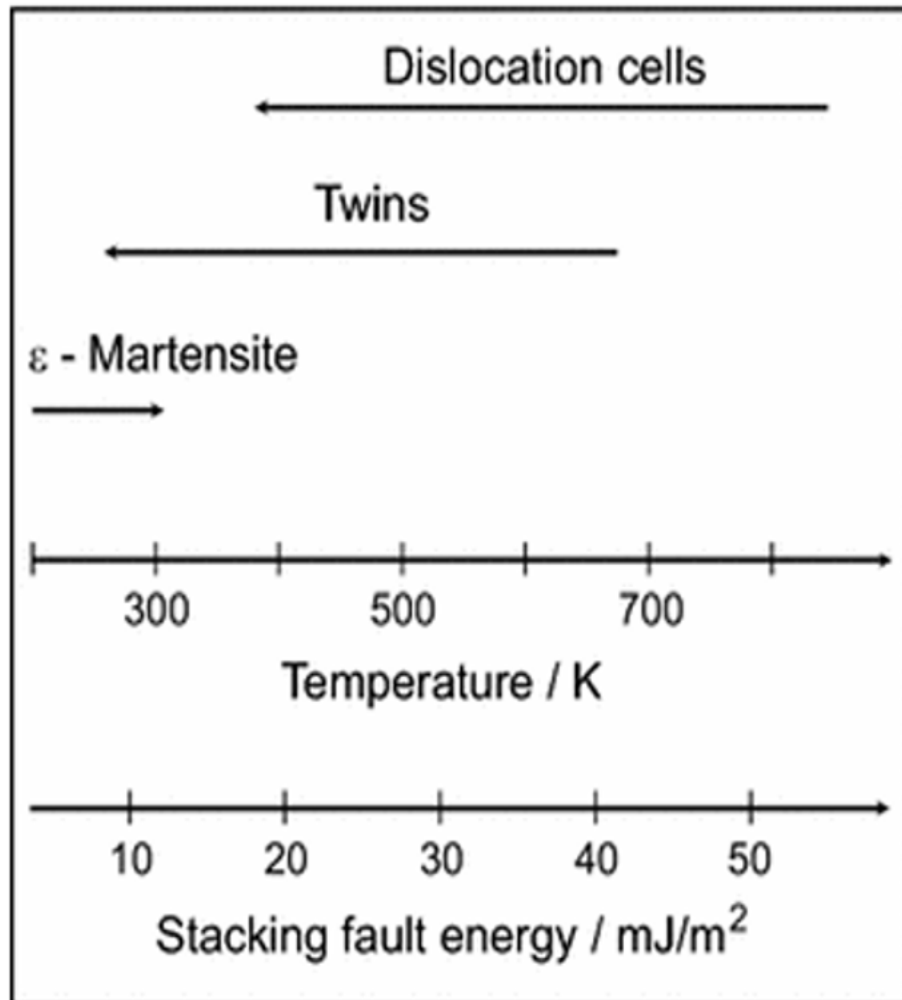
Fe (%) (balance)	79.50	Cr (%)		Mn (%)	20
Ni (%)		Al (%)		Si (%)	
C (%)	0.5	Cu (%)		N (%)	

Temperature

T (K) (optional, default 293)



SFE vs plastic deformation mechanisms



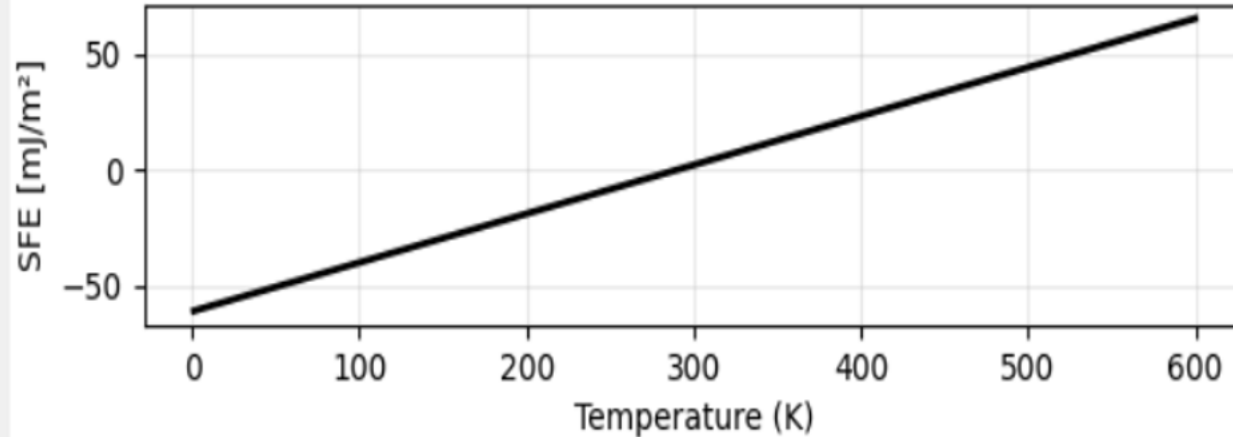
A representative plot for the Fe-20Mn-0.5C system

Plots generated. Room temp (293K) SFE: 23.86 mJ/m². The predicted plastic deformation mechanism is Twins.

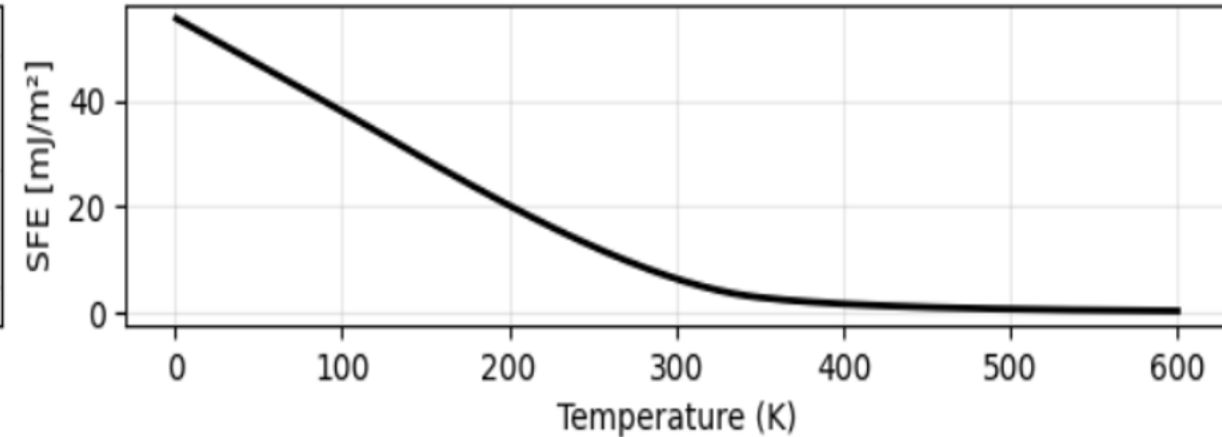
Deformation
mechanism

SFE Analysis Plots

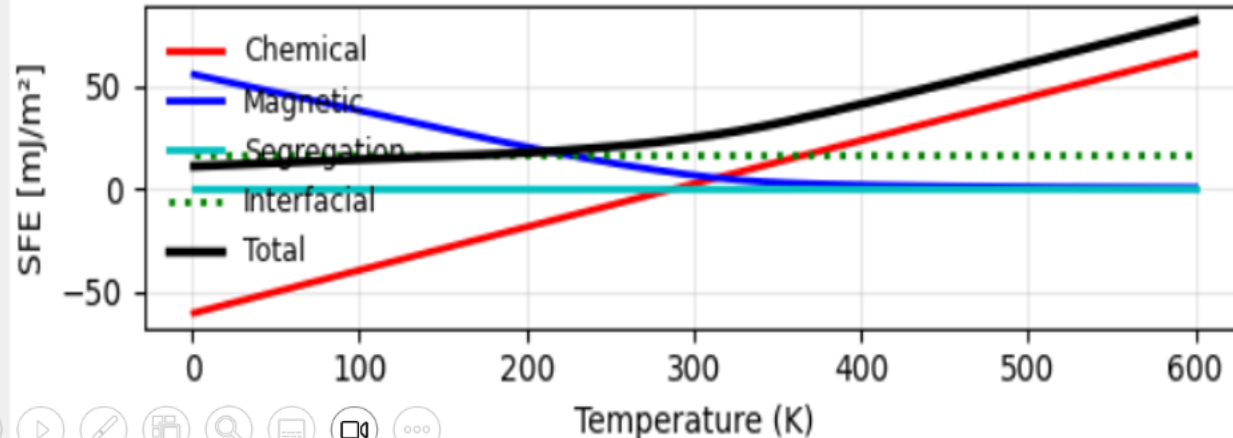
Chemical SFE Contribution



Magnetic SFE Contribution



SFE and Components



SFE vs. Temperature

